

AI CONTRACT RISK ANALYZER

D0. Project foundation and scope

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Project idea

The **AI Contract Risk Analyzer** is an intelligent web-based platform that helps long-term freelancers, contract-based commercial talents, small business owners, agency founders, and corporate operations managers review legal contracts, detect high-risk clauses, estimate potential financial impact, and generate ready-to-send negotiation messages before signing.

Personas carried forward from Lab 2

The project uses four primary personas from the submitted Lab 2. Their goals are different, but they interact with the same platform capabilities, so they are represented by one primary **Contract User** actor.

Persona	Main goal	Main pain point
Sara Al Harbi	Protect payment milestones, revision limits and conditional IP transfer on single gigs.	Dense legal wording, pressure to sign quickly, and unclear counter-offer language.
Omar Al Qahtani	Prioritize supplier-contract risks by estimated financial exposure and buyer perspective.	Surprise fees, renewal clauses, early-exit penalties and limited legal time.
Tariq Al Mansoor	Detect uncapped liability and IP exposure and export a structured audit report.	Slow manual review and legal turnaround delays during enterprise deals.
Noura Al Ghamdi	Triage routine vendor agreements and escalate only genuine legal risks.	Manual review and legal backlog delay operational rollouts.

Actor model

Actor	Description and goals	Challenges / notes
Contract User	Uploads contracts, selects the relevant role, reviews risk findings, checks financial exposure, generates negotiation content, compares agreements, and exports reports.	Includes the four Lab 2 personas; the UI must remain usable for both medium- and high-technical-skill users.
AI Analysis Service	External AI service invoked to extract/interpret contract content and identify risky clauses.	Service output must be treated as analysis support rather than legal advice.
PDF/Text Extraction Service	Supporting service that converts uploaded PDF content into analyzable text.	Poor scans or unusual formatting can reduce extraction quality.

Scope and design assumptions

- The MVP is a web-based contract review platform focused on pre-signing analysis.
- The system analyzes uploaded legal contracts and presents clause-level risks in plain language.
- Role perspective (for example, Buyer or Seller) is used to tailor recommendations.
- Financial exposure is an estimated analytical value derived from contract terms, not a guaranteed legal or financial result.
- Generated negotiation wording and reports are decision-support outputs; the platform does not replace a qualified lawyer.

Warm-up: actions and entities

Actions: upload, analyze, select role, review, rank, compare, detect, estimate, generate, configure and export.

Entities: User Profile, Contract, Clause, Risk Finding, Negotiation Message and Audit Report.

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D1. Prioritized use cases and traceability

ID / use case	Primary actor	Goal / outcome	Priority	Req.
UC-1 Analyze Contract	Contract User	Upload a contract, select the role, and receive a clause-level risk analysis.	M	R1,R4,R7,R10,R11
UC-2 Review Risk Findings	Contract User	Understand flagged clauses, severity and rationale in plain language.	M	R1,R7,R10
UC-3 Estimate Financial Exposure	Contract User	Rank clauses by estimated money at stake.	M	R5
UC-4 Generate Negotiation Message	Contract User	Produce diplomatic substitute wording and a ready-to-send message.	M	R2,R3
UC-5 Compare Agreements	Contract User	Compare clauses and risk findings across multiple agreements.	S	R6
UC-6 Export Audit Report	Contract User	Export a structured contract risk audit summary.	M	R8
UC-7 Configure Risk Thresholds	Contract User	Set agency-specific risk thresholds and standards.	S	R9
UC-8 Batch Triage Contracts	Contract User	Upload several contracts and rapidly identify agreements requiring escalation.	M	R11
UC-9 Generate Legal Escalation Summary	Contract User	Produce a concise summary containing highlighted flagged clauses for legal review.	M	R12

MoSCoW priority

M = Must have **S = Should have**

Must-have scope covers contract analysis, risk review, financial exposure, negotiation assistance, exportable reporting, and batch triage. Comparison and custom agency thresholds are treated as should-have features for the initial scope.

Requirements traceability matrix

Req.	Requirement / persona evidence	Use cases
R1	Plain-language clause risk explanations and visual severity indicators; Sara.	UC-1, UC-2
R2	Safer clause wording and ready-to-send negotiation messages; Sara.	UC-4
R3	Single-gig conditional IP transfer checking until full payment; Sara.	UC-1, UC-4
R4	Counterparty role perspective (Buyer vs. Seller); Omar.	UC-1
R5	Financial exposure ranking by money at stake; Omar.	UC-3
R6	Multi-agreement clause comparison dashboard; Omar.	UC-5
R7	Uncapped liability and mutual indemnification risk detection; Tariq.	UC-1, UC-2
R8	Exportable contract risk audit summary report (PDF / Word); Tariq.	UC-6
R9	Custom risk threshold configuration for agency standards; Tariq.	UC-7
R10	Automated SLA compliance checklist and non-standard clause detector; Noura.	UC-1, UC-2
R11	Batch document upload and rapid triage risk dashboard; Noura.	UC-1, UC-8
R12	Legal escalation summary report with highlighted flagged clauses; Noura.	UC-9

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D2. Fully dressed use case specifications

UC-1 Analyze Contract

Field	Specification
Scope / level	AI Contract Risk Analyzer / user goal
Primary actor / trigger	Contract User / requests analysis for a contract.
Stakeholders	User wants fast, understandable, actionable risk information before signing.
Preconditions (input)	A readable contract file is selected; the user provides the relevant counterparty role.
Success guarantee (output)	The contract is analyzed and a structured set of risk findings is shown with severity, rationale and estimated exposure where available.
Minimal guarantee	Invalid input does not create a partial analysis or misleading result; the original uploaded contract remains unchanged.

Main success scenario

1. User requests **Analyze Contract**.
2. System requests the contract file and counterparty role.
3. User uploads a readable contract and chooses the relevant role.
4. System extracts the contract text and validates that content is available.
5. System analyzes clauses and identifies potential risk patterns.
6. System ranks findings using severity and estimated financial exposure where applicable.
7. System presents a plain-language summary and detailed flagged clauses.
8. System offers next actions such as negotiation wording, report export, comparison, or escalation summary.

Extensions

- **3a. Unsupported file / unreadable scan:** show a clear extraction error and request another file.
- **4a. Empty or incomplete extracted text:** stop analysis and explain the problem.
- **5a. AI analysis service unavailable:** preserve the uploaded file and allow retry without duplicating the contract.
- **6a. Financial value cannot be estimated:** show the qualitative risk without inventing a monetary value.

Special requirements and constraints

The interface should make high-risk clauses easy to scan, support the persona's role perspective, and clearly distinguish analytical recommendations from legal advice.

UC-9 Generate Legal Escalation Summary

Field	Specification
Scope / level	AI Contract Risk Analyzer / supporting system goal
Primary actor / trigger	Contract User / requests escalation summary after risk review.
Stakeholders	Operations manager or agency leadership wants legal counsel to focus on genuine risks.
Preconditions (input)	A completed analysis exists and one or more findings are flagged for escalation.
Success guarantee (output)	A concise summary is generated with the contract name, highlighted risky terms, severity and reasons for legal review.
Minimal guarantee	If report generation fails, no contract or analysis data is lost and the user can retry.

Main success scenario

1. User requests an escalation summary.
2. System loads the completed risk analysis.
3. System selects flagged clauses that meet the escalation threshold.
4. System summarizes each flagged clause in concise business language.
5. System creates the escalation report with the relevant clause references.
6. System presents the report for review and export.

Extensions

- **3a. No clause meets the threshold:** inform the user that no escalation is currently required.
- **4a. A finding is incomplete:** exclude it from the report and show that it requires manual review.
- **5a. Export fails:** keep the report available in the current session and offer retry.

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D3. Use case diagram

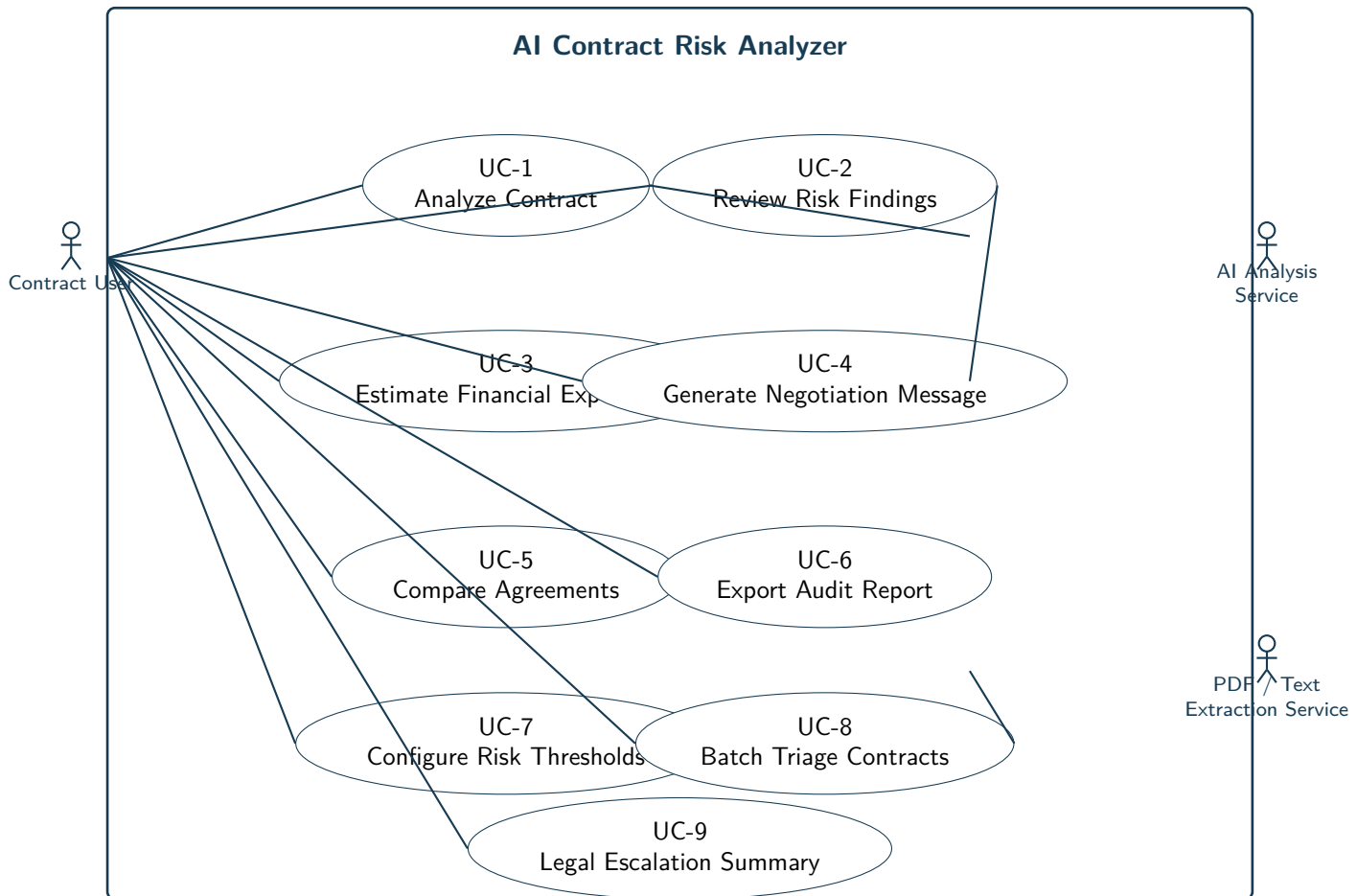
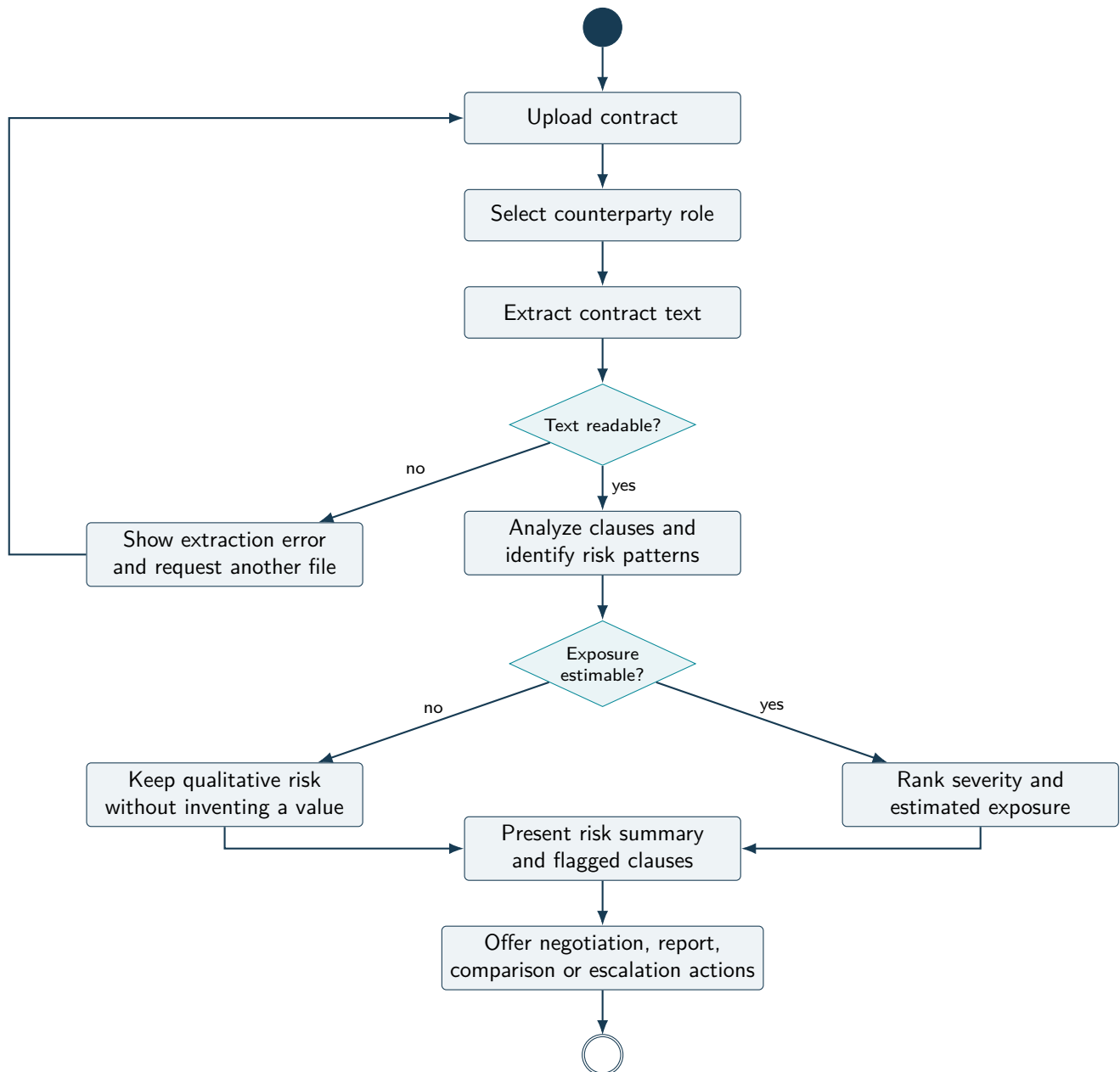


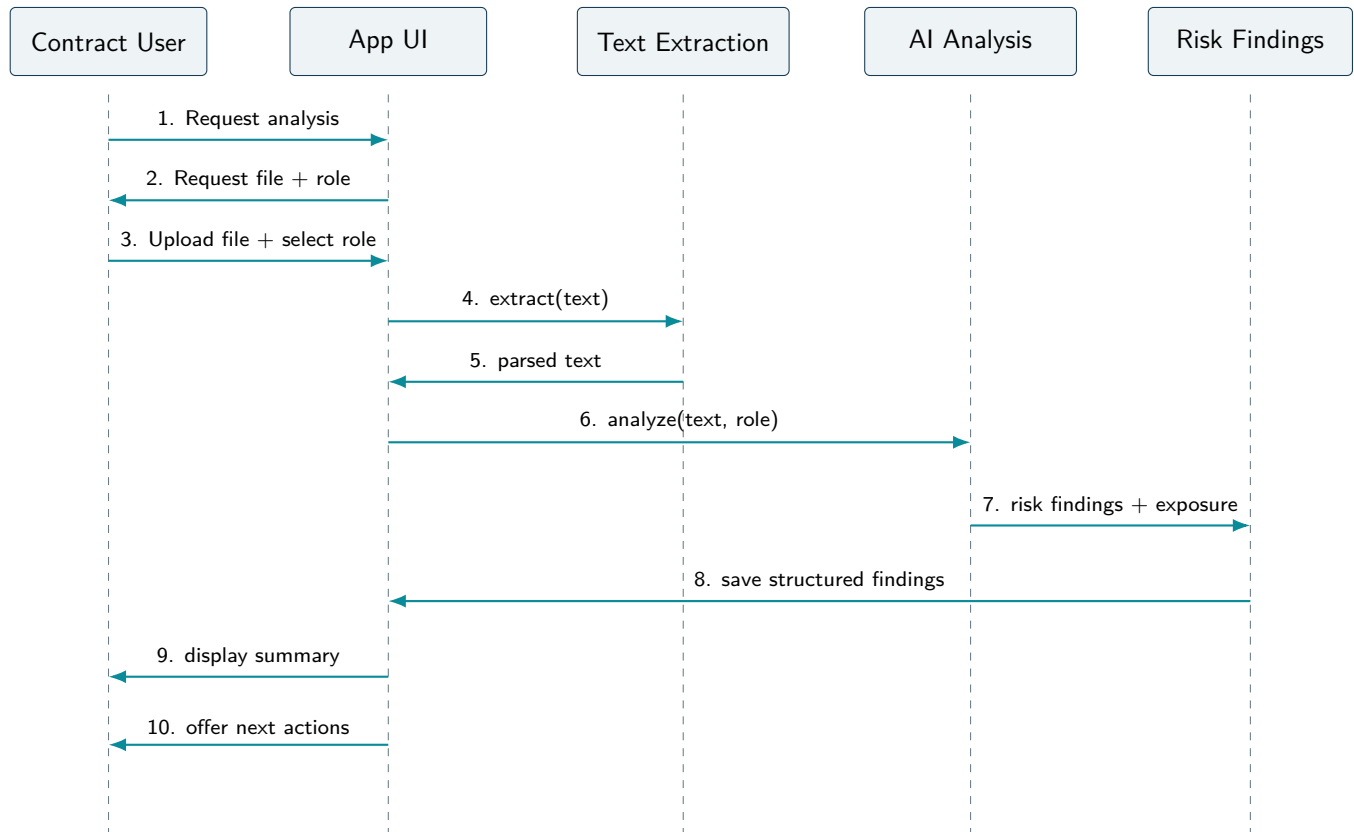
Diagram note. The **Contract User** represents all four Lab 2 personas because their goal is contract risk review and their permissions are equivalent. The AI and extraction services are supporting external systems; the user-facing use cases remain goal-level.

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D4. Activity diagram: UC-1 Analyze Contract

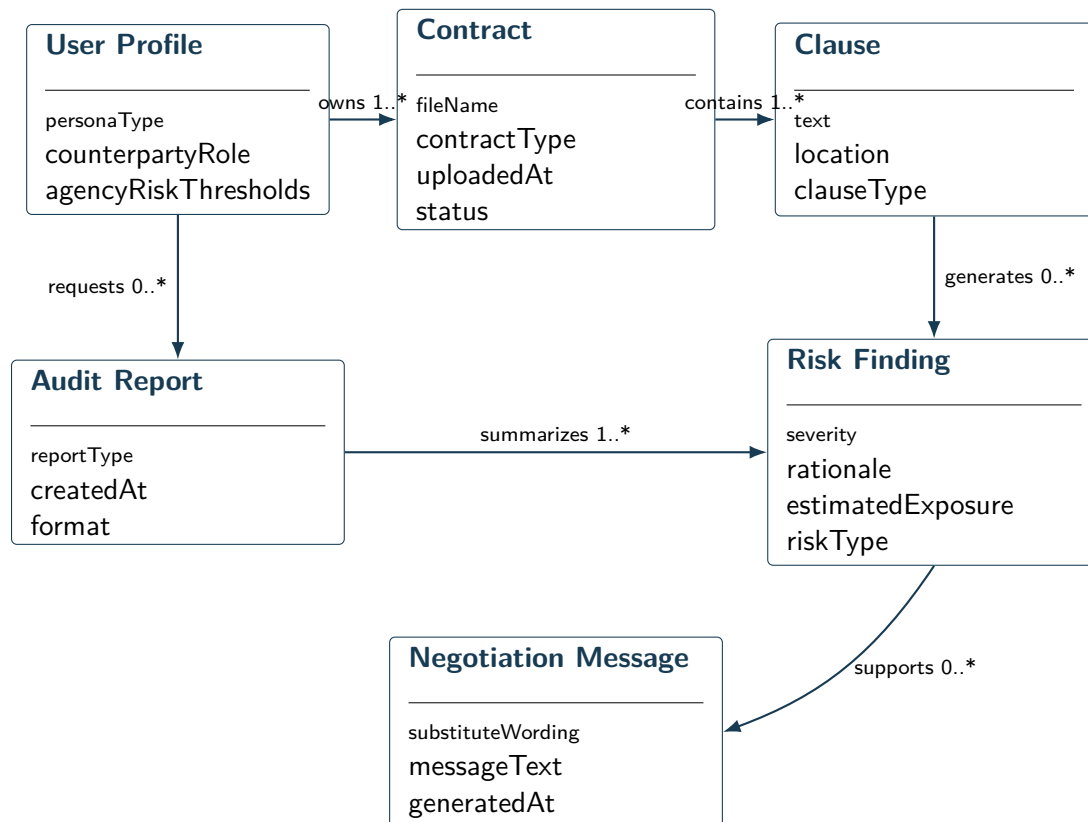
Behavior covered: valid input, unreadable documents, analysis, uncertain monetary exposure, and presentation of actionable findings. The branches reflect the main extensions in UC-1.

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D5. Sequence diagram: UC-1 success

Sequence alignment. Steps 1-10 follow the success path of UC-1: request, input, extraction, AI analysis, structured risk findings, and user-facing results.

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D6. Domain model (class diagram)**Domain rules and multiplicities**

- Each Contract belongs to exactly one User Profile; a profile can own zero or many contracts.
- Each Contract contains one or more analyzed Clauses after successful extraction.
- Each Clause can produce zero or many Risk Findings because one clause may trigger more than one risk category.
- A Risk Finding can support zero or many Negotiation Messages, while messages are generated only from available findings.
- An Audit Report summarizes one or more Risk Findings and is associated with the user requesting the report.
- The model remains conceptual: it intentionally avoids UI, database, API and framework-specific classes.

Consistency checklist

Check	Evidence / result
Actors and traceability	All four Lab 2 personas map to Contract User; supporting AI and extraction services are explicit.
Use case coverage	All prioritized goal-level use cases appear in D3 and are traced to R1-R12.
Relationships	Associations use clear goal-level links; no unnecessary include/generalization is introduced.
Sequence alignment	D5 follows the successful UC-1 flow from upload through extraction, analysis and results.
Domain coverage	D6 represents the main business concepts used by analysis, negotiation and reporting.
Scope consistency	D0 assumptions, D1 requirements, D2 narratives and D3-D6 diagrams use the same terminology.

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D7. Peer-review sheet

Reviewing team **AI Game Analyzer (Aziz and Saif)**
Reviewer names Aziz and Saif
Review date Fall 2026

Review criterion	Pass	Comments
Actors cover all personas and external systems.	✓	Pass
Use case names are goal-level and clear.	✓	Pass
Includes / extends are used correctly.	✓	Pass
Activity and sequence diagrams match narratives.	✓	Pass
Domain model has valid concepts and multiplicities.	✓	Pass
Preconditions and success guarantees are clear.	✓	Pass
Names and behavior are consistent across artifacts.	✓	Pass
Layout, clarity and readability are suitable.	✓	Pass

Review outcome

Final reviewer decision: ✓ Pass / Ready for submission

Strengths: Clear persona-to-actor mapping, traceable requirements, aligned UML artifacts, and readable goal-level use cases.

Issues found: None recorded.

Required revisions: None.